Lead-lag design for a normalized camera axis

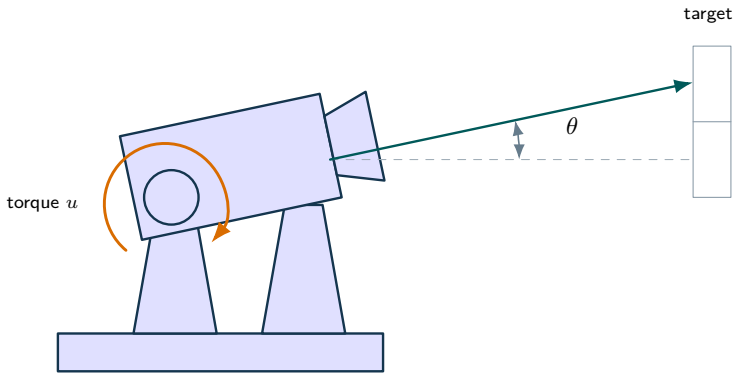
Original learning collection – November 2026

Evidence note. Original teaching scenario. Numerical parameters are deliberately chosen for learning. All generated data and curves are synthetic; no physical measurement is asserted.

1 Design brief

For $G(s) = 1/[s(s+1)]$, design lead compensation with crossover $\omega_c = 3\,\text{rad/s}$ and target phase margin 55°, centering maximum lead at crossover. Add lag zero $0.03\,\text{rad/s}$ and pole $0.006\,\text{rad/s}$, then renormalize gain at ω_c. Compare final phase margins, velocity constants K_v, ramp errors and reference responses.

2 Physical system



Gimbal axis with angular readout

Normalized model $G(s) = 1/[s(s+1)]$; geometry is illustrative

A camera gimbal illustrates angular tracking and disturbance rejection. Its normalized plant is stipulated, with no claim that geometry alone determines its two poles.

3 Lead synthesis

At $\omega_c = 3$, the plant phase is $-90° - \tan^{-1}(3)$. Required additional phase is $\phi = 55° - 180° - \angle G(j3)$, a positive value less than 90 degrees. For

$$C_a = K\frac{Ts+1}{\alpha Ts+1}, \quad \alpha = \frac{1-\sin\phi}{1+\sin\phi}, \quad T = \frac{1}{\omega_c\sqrt{\alpha}},$$

the maximum phase lead occurs at the prescribed crossover. Its magnitude there is $K/\sqrt{\alpha}$, giving $K = \sqrt{\alpha}/|G(j\omega_c)|$.

4 Lag extension

Use $C_b = (1 + s/0.03)/(1 + s/0.006)$. Its DC gain is one and high-frequency gain is 0.2. Renormalize with $\gamma = 1/|C_a(j3)C_b(j3)G(j3)|$. The combined controller is $\gamma C_a C_b$. The lag decreases the margin slightly; it does not preserve the lead-only margin exactly. Because its corner frequencies are far below crossover, the reduction is small while the low-frequency loop gain rises by approximately five.

5 Ramp accuracy

For this type-one plant, $K_v = \lim_{s\to 0} sC(s)G(s) = C(0)$ and the unit-ramp error is $1/K_v$, provided the closed loop is stable. Reference-step final values remain one, so a step alone cannot reveal the improvement in ramp accuracy.

6 Tradeoffs

The slower lag dynamics can extend settling and store controller state. This normalized exercise does not include actuator saturation, camera inertia units, backlash or sensor noise; the controller is not a hardware calibration.

7 What to learn from the experiment

A meaningful compensation comparison needs both stability margin and low-frequency accuracy. Keep the slow lag transient visible when interpreting ramp error.

8 System sketch

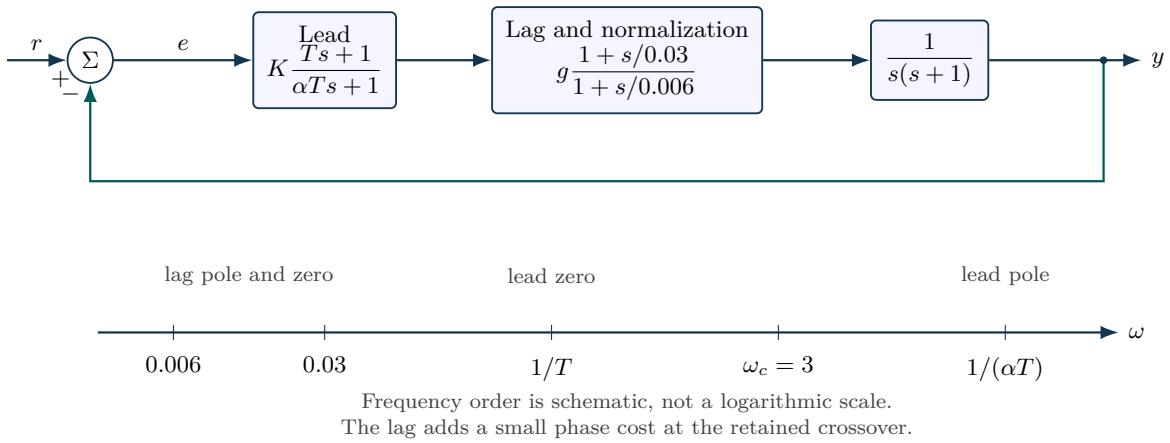
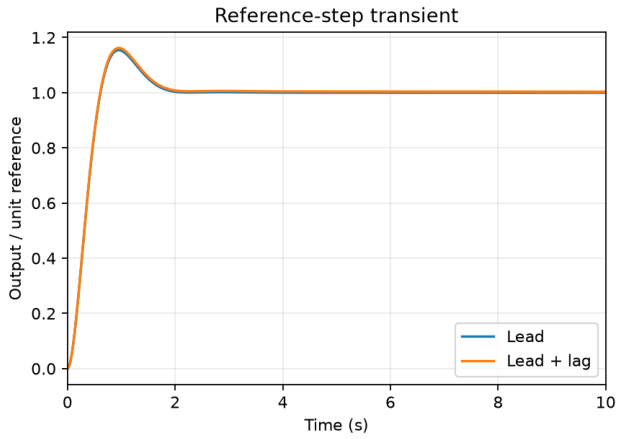
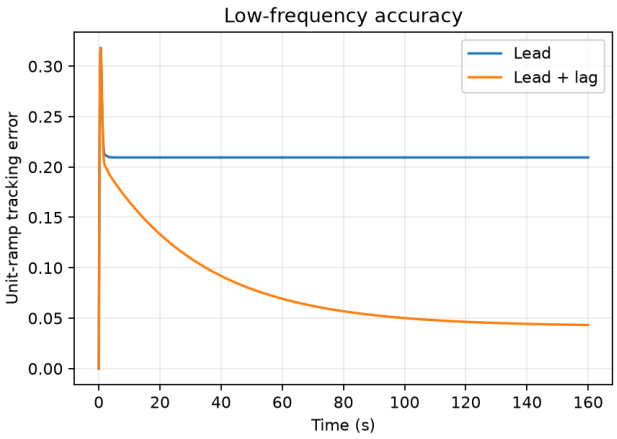


Figure 1. Lead and lag sections are drawn separately so their gain normalization and frequency ordering can be analyzed independently.

9 Executed computational evidence

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Lead: phase margin=55.000000 deg, Kv=4.775006, limiting unit-ramp error=0.209424
Lead + lag: phase margin=54.541653 deg, Kv=23.873883, limiting unit-ramp error=0.041887
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10 Method references

The equations use standard balance laws and feedback-control methods. Further explanation: https://apmonitor.com/pdc/ and https://fbsbook.org/. Implementation documentation: https://python-control.readthedocs.io/en/0.10.2/. See the repository reference catalog for individual topics and videos.